

Network security threats are attacks that try to affect the availability, confidentiality, or integrity of network systems and data. Some attacks are mainly focused on taking a service offline, while others try to intercept communication or hide the real source of malicious traffic.

1. DoS and DDoS Attacks

A **Denial-of-Service (DoS)** attack attempts to make a system, server, or network service unavailable to legitimate users.

The basic idea is simple: the attacker sends a large amount of traffic or requests to the target until its resources are exhausted. The server may become very slow or completely unavailable.

What is a DDoS Attack?

A **Distributed Denial-of-Service (DDoS)** attack has the same general goal, but the traffic comes from many different compromised devices instead of one source.

Attackers can use a **botnet**, which is a group of infected devices controlled by the attacker. Because the traffic comes from many sources, DDoS attacks can be much harder to block.

How the Attack Works

A typical DDoS attack can follow these steps:

1. The attacker compromises or controls a large number of devices.
2. These devices become part of a botnet.
3. The attacker sends commands to the infected devices.
4. The devices send large amounts of traffic or requests toward the target.
5. The target consumes its available bandwidth, CPU, memory, or connection resources.
6. Legitimate users experience slow performance or cannot access the service.

Real-World Example

One well-known example is the **Mirai botnet attack in 2016**. Mirai infected vulnerable IoT devices such as cameras and routers and used them to generate large-scale DDoS traffic.

The attack affected Dyn, a major DNS infrastructure provider, and caused access problems for a number of popular websites.

Possible Impact

DDoS attacks can cause:

- Website or application downtime
- Slow network performance
- Loss of revenue
- Increased infrastructure costs
- Damage to the organization's reputation
- Disruption of important business services

Mitigation

Three important ways to reduce DDoS risk are:

1. DDoS Protection Services

Organizations can use specialized DDoS protection and traffic-filtering services to detect and absorb malicious traffic.

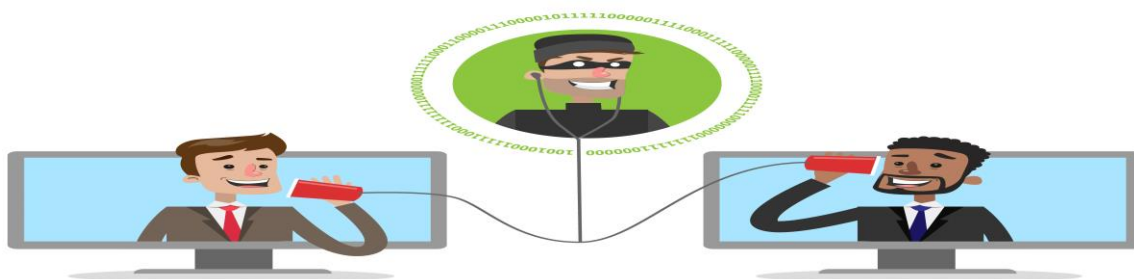
2. Rate Limiting

Rate limiting can restrict how many requests a user or IP address can send within a specific period. This can help reduce the effect of request-based attacks.

3. Network Monitoring

Continuous monitoring can help security teams identify unusual traffic spikes and respond before the service becomes completely unavailable.

2. Man-in-the-Middle (MITM) Attack



What is a MITM Attack?

A **Man-in-the-Middle attack** happens when an attacker secretly places themselves between two parties communicating with each other.

The attacker may be able to monitor, modify, or redirect the communication without the victims realizing that someone is intercepting the connection.

How the Attack Works

A simplified MITM attack can happen as follows:

1. Two users or systems start communicating.
2. The attacker finds a way to position themselves between the two parties.
3. Traffic passes through the attacker's system.
4. The attacker can observe or potentially modify the communication.
5. The two parties may continue communicating without knowing that their traffic has been intercepted.

One common example is an attacker creating a fake Wi-Fi network that looks legitimate. A victim connects to it and the attacker can attempt to observe the victim's network traffic.

Real-World Example

MITM attacks have commonly been associated with attacks on users connected to **unsecured or malicious public Wi-Fi networks**.

For example, an attacker could create a Wi-Fi network with a name similar to a legitimate network in a public location. If a user connects to the fake network, the attacker may attempt to intercept traffic or redirect the user to malicious websites.

Possible Impact

A successful MITM attack can lead to:

- Theft of login credentials
- Exposure of sensitive information
- Session hijacking
- Modification of transmitted data
- Privacy violations
- Redirection to malicious websites

Mitigation

1. Use HTTPS

HTTPS encrypts communication between the client and the web server using TLS, making it much harder for an attacker to read or modify the traffic.

2. Avoid Untrusted Networks

Users should avoid connecting to unknown or suspicious Wi-Fi networks, especially when handling sensitive information.

3. Use VPNs When Appropriate

A trusted VPN can provide an additional layer of protection by encrypting network traffic between the device and the VPN server.

3. IP Spoofing

What is IP Spoofing?

IP Spoofing is a technique where an attacker changes the source IP address in network packets to make the traffic appear to come from another IP address.

The attacker is essentially hiding the real source of the traffic or pretending to be another system.

How the Attack Works

A simplified IP spoofing process is:

1. The attacker creates a network packet.
2. The attacker changes the source IP address in the packet.
3. The packet is sent toward the target.

4. The target sees the spoofed IP address instead of the attacker's real address.
5. Depending on the attack, the spoofed traffic may be used to bypass weak filtering or support another attack.

IP spoofing is especially useful in attacks where the attacker does not need to receive a direct response from the target.

Real-World Example

IP spoofing is commonly used as part of **DDoS attacks**, especially reflection and amplification attacks.

For example, an attacker can send requests to third-party servers while using the victim's IP address as the spoofed source. Those servers may then send their responses to the victim, creating a much larger amount of traffic toward the victim.

Possible Impact

IP spoofing can contribute to:

- DDoS attacks
- Bypassing weak IP-based access controls
- Hiding the attacker's actual IP address
- Network reconnaissance
- Reflection and amplification attacks

Mitigation

1. Ingress and Egress Filtering

Network administrators can configure filtering rules to block packets with source addresses that should not appear on a particular network.

2. Strong Authentication

Organizations should not rely only on IP addresses for authentication or authorization. Strong authentication mechanisms provide better protection.

3. Network Monitoring

Monitoring unusual traffic patterns and unexpected source addresses can help identify spoofing-related activity.

Comparison of the Attacks

Attack	Main Attack Vector	What is at Risk?	Difficulty	Ease of Mitigation
DoS/DDoS	Large amount of malicious traffic or requests	Availability	Medium	Medium
MITM	Intercepting network communication	Confidentiality and Integrity	Medium to High	Medium
IP Spoofing	Forged source IP addresses	Trust, availability, and network controls	Medium	Medium

The three attacks are different in their main objectives. DDoS mainly targets **availability**, MITM attacks mainly threaten **confidentiality and integrity**, while IP spoofing is often used to hide the attacker's identity or support other attacks.

Conclusion

After reviewing these threats, there are three important points to remember:

1. **No single security control is enough.** Network security requires multiple layers such as firewalls, monitoring, encryption, authentication, and filtering.
2. **Encryption is very important for protecting communication.** Using HTTPS and TLS can significantly reduce the risk of attackers reading or modifying network traffic.
3. **Monitoring and early detection matter.** Unusual traffic patterns, unexpected connections, and sudden increases in network activity can be useful indicators of an attack.

Understanding how these attacks work helps security teams choose the right defensive controls and respond to incidents more effectively.

References

1. National Institute of Standards and Technology (NIST) – Network Security and Cybersecurity guidance.

<https://www.nist.gov/>

2. Cybersecurity and Infrastructure Security Agency (CISA) – Distributed Denial-of-Service guidance.

<https://www.cisa.gov/>

3. OWASP – Transport Layer Security and web application security guidance.

<https://owasp.org/>

4. Cloudflare Learning Center – DDoS, MITM, and IP spoofing explanations.

<https://www.cloudflare.com/learning/>